

Hotel Performance Analysis

Executive Summary | Corporate Analyst Portfolio Project

Key Performance Indicators

Metric	Value	Insight
Total Bookings Analyzed	119,208	2015–2017 dataset
Overall Cancellation Rate	37.1%	Industry avg ~20% — elevated
Average Daily Rate (ADR)	\$100.17	Confirmed stays only
Estimated Total Revenue	\$26.0M	Confirmed stays only
Average Length of Stay	3.4 nights	Both hotel types combined
Average Booking Lead Time	104 days	Higher lead time → higher cancel risk

Cancellation Risk Analysis

At 37.1%, the overall cancellation rate significantly exceeds the hospitality industry average of ~20%. Segment-level analysis exposes where the risk is concentrated:

- **Undefined:** 100.0% cancellation rate (2 of 2 bookings cancelled)
- **Groups:** 61.1% cancellation rate (12,094 of 19,790 bookings cancelled)
- **Online TA:** 36.8% cancellation rate (20,735 of 56,408 bookings cancelled)
- **Offline TA/TO:** 34.3% cancellation rate (8,301 of 24,181 bookings cancelled)

Recommendation: Introduce targeted deposit policies for the Online TA and Groups segments. A non-refundable tier with a modest discount incentive could materially reduce revenue leakage from late cancellations.

Revenue & Distribution Channel Performance

Direct bookings carry a 17.5% cancellation rate and higher ADR — less than half the TA/TO channel rate of 41.1%. Shifting channel mix toward Direct improves both margin and forecast reliability.

Channel	Confirmed	Avg ADR	Revenue	Cancel Rate
TA/TO	57,614	\$101.78	\$20.8M	41.1%
Direct	12,055	\$106.30	\$4.3M	17.5%
Corporate	5,184	\$67.49	\$0.8M	22.1%
GDS	156	\$119.93	\$0.0M	19.2%

Undefined	1	\$112.70	\$0.0M	80.0%
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Forecast Accuracy & Recommendations

- Segment forecasts separately — Online TA and Groups need a cancellation haircut of 35–99%
- Apply lead-time decay factors: bookings 90+ days out cancel at nearly 2× the rate of same-week bookings
- Track rolling 30/60/90-day cancellation velocity as a leading indicator
- Treat Non-Refundable deposit bookings as near-certain revenue — they cancel at <1%

Analysis based on 119,208 hotel bookings (2015–2017) | Source: Hotel Booking Demand Dataset (Kaggle) | Tools: Python · pandas · SQL · Power BI